

Ohio State University, Prof. Baliga at North Carolina State University, Prof. DeDoncker at University of Aachen (Germany), to name a few. I also greatly appreciate my mentor Dr N.G. Hingorani, father of power electronics innovations and applications, who has been a great teacher and inspiration for the last 38 years and taught me humility as well as pragmatism.

Have you been initiating such tie-ups in the US too, or was it something you tried in India for the first time? If in US too, how was the Indian experience different?

In the US, both Silicon Power and SiCmore Semi have been working with Ohio State, NC State, and RPI, to mention a few. We lean on these organisations for materials, devices, packaging, and system development. We expect similar relationships in India as well. I truly believe that being born in India has greatly helped and enhanced my overall experience with my personal as well as professional journey. I strongly believe in and have ingrained in my life two eternal and core teachings of the Gita—Do your best and do not worry about the results! I also blindly believe that whatever happens is good for the long term.

Both my experiences in the US and India have been rewarding as we got to work with some of the sharpest minds in academia. We hope, over time, to be able to hire more students from India's institutions.

Going back to the business model, do you see any challenges in creating the material?

The challenges are going to be there. With the start of anything new, its qualification as a process itself is a big challenge. Even though the semiconductor processes are modern and detailed, challenges exist there too. You have to keep debugging how small things work in a new environment.

Things are also quite dependent on humidity, temperature conditions, the environment, cleanliness, and a lot of other factors. Those processes take time to perfect in a new environment. We have been planning for the same and are sure that we will be able to get over it.

The other challenge that we may come across is the availability of things like chemicals in as pure a state as we need in India. We might have to import such materials in that case. With chemicals, there is a lot of planning needed

as they are not allowed to be imported through air. It becomes necessary to plan intelligently in all such cases as we have to make sure to have lead-time.

By when do you plan to start the packaging part?

Joint secretary Gaur inaugurated the clean room only a few days ago. We already have a shell in Halol that is ready to accept the equipment. The epi system will arrive in about two-to-three months from now and we will then start putting everything together. I would say packaging equipment would start to arrive in about six months from now.

Do you think the 'not so pro-China' mood will help you in any manner?

Certainly! The geo-political conditions that have developed over the last one or two years will help us. I call them tail winds. While the tail winds will help us propel forward, they are not primary factors. I am sure that even without these tail winds, we will be able to fly successfully.

Is it possible that you will also be offering services to high-quantity players?

We would provide value-added services, materials, packaging and so forth. Where we feel we may not be able to go after is very high-volume, relatively lower-power silicon carbide chips which other companies are doing through billion-dollar equipment right now.

Which companies would you be typically competing against in this case?

CREE, STMicroelectronics, and the likes of them. They are high-volume ones. Whenever a new entity comes, the bigger companies, whether intentionally or unintentionally, try to cut prices publicly. This is done to diminish the progress of the new company. As we go high-power, our first few steps would be lower-power, which will compete with them.

Our strategy would be to not compete head-to-head but provide some smart intelligent solutions to the customers that would use our products.

Does that mean that you are looking to provide customers with complete solutions including reference designs?

Absolutely! Silicon Power is a vertically-integrated company today. Not only do we design our own components but also manufacture them.

Silicon Power also packages them and provides a full system that ultimately goes into the end use. We have been installing complete power-management systems back in the US.

Do your plans include gallium nitride components?

What we plan to do is look at a technology and a product from the technical as well as the economic viability angles. When you look at the semiconductor sector, silicon material has matured over the last fifty years. The next was silicon carbide, and that is the reason it is far ahead in terms of the material and maturing technology in comparison to gallium nitride.

The latter is not there yet; you want to make sure that instead of going after one defect-free wafer, you go after manufacturable and reliable product supply chains. The second thing is when it comes to the applications, silicon carbide, in terms of performance, meets all the criteria that high-performance systems generally require. Though people are trying to push gallium nitride for power convertors, I do not see that happening, unless some big breakthrough happens.

Do you find the recent policy announcements to be helpful?

The government policies introduced in the last eighteen months are very industry-friendly. These are very pro-growth. I think they want to make 'Make in India' as friendly as it can be. I understand that the government has a different set of organisations that execute the whole policy. Yes, we are very happy with Ministry of Electronics and Information Technology (MeitY), Invest in India, and others, but to work with IFCI (the erstwhile Industrial Finance Corporation of India Ltd) on the project side, you have to learn from your mistakes and implement the learnings.

Who would you want to reach out to through this interview?

One of the most important sectors that can help us is the customers. They are buying these products and buying from outside India. We have already approached a lot of big companies working in the electric vehicle space in India. Our approach is to tell these companies that let us try to work together and develop the most optimum product for your application. This would ensure that it is not merely a customer vendor thing but a strategic relationship. **EFY**